

A057 v0.1.0: Master-Mass Reconstruction Blind Falsifier

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1 Identification

Catalog ID A057; release v0.1.0; SST Falsifier Framework v1.0.6, profile `multilibrary_gpu`. The framework protocol is fail-closed, whereas scientific gates use diagnostic continuation.

2 Question

Does a topology-stable finite-core vortex energy and its variational spectrum reconstruct a compact mass kernel before protected target comparison?

3 Hypotheses

The null states that independently sourced carriers do not exhibit a reproducible topology-level energy/dynamical kernel beyond numerical and source variation. The alternative requires reproducible topology structure and protected-holdout generalization before any target reveal.

4 Assumptions

The tested sector is incompressible, inviscid, closed-filament dynamics without reconnection or external forcing. E010/PKLSA is geometry/provenance authority, not particle authority. The physical reference object is the three-dimensional kinetic-energy integral. The operational primary kernel is a regularized slender-filament Hamiltonian; an independent Bishop-frame core-tube quadrature is retained as a second reduced finite-core diagnostic.

5 Symbols

$\mathbf{X}_i(s)$ denotes a closed filament component, a the regularization scale, Γ circulation, H_a the line Hamiltonian, q reduced Bishop-frame coordinates, K_H the energy Hessian and J the first-order Jacobian.

6 Equations

Machine-readable science contract

Research question Does a source-independent, topology-stable finite-core vortex Hamiltonian and its local variational spectrum provide a lower-complexity reconstruction of the historical SST mass kernel, with frozen blind mass-scale predictions that generalize across independent PKLSA carriers?

Objective Compute geometry, regularized filament energy, interaction coherence, reduced Lagrangian/Hamiltonian closure, Hessian/Jacobian spectra, and target-free surrogate competition on source-native PKLSA carriers. Freeze all topology-level kernels and anonymous

composite predictions before any protected target comparison. Scientific gate failures are diagnostic and do not suppress execution of later scientific gates; only protocol/integrity failures are hard blockers.

H_0 After numerical convergence and source-independence controls, no registered energy or spectral functional is stable within topology relative to between-topology variation, and added dynamical descriptors do not improve protected holdout prediction of the finite-core energy kernel beyond simple geometry controls.

H_1 At least one preregistered finite-core energy/dynamical functional is reproducible across independent carriers of the same topology, separates topology classes beyond numerical/source variation, and yields frozen target-free predictions whose post-reveal mass ratios outperform the registered historical/factorized baselines without target-specific fitting.

Registered equations

E1: Canonical local mass-equivalent density scale

$$\rho_m = \frac{\rho_f \|\mathbf{v}_\odot\|^2}{2c^2} \quad (1)$$

Dimensional check: $(\text{kg m}^{-3})(\text{m}^2 \text{s}^{-2})/(\text{m}^2 \text{s}^{-2}) = \text{kg m}^{-3}$.

E2: Regularized finite-core filament Hamiltonian

$$H_a = \frac{\rho \Gamma^2}{8\pi} \sum_{i,j} \oint_{K_i} \oint_{K_j} \frac{d\mathbf{X}_i \cdot d\mathbf{X}_j}{\sqrt{\|\mathbf{X}_i - \mathbf{X}_j\|^2 + a^2}} \quad (2)$$

Dimensional check: $\rho \Gamma^2 \text{ times a length has units } (\text{kg m}^{-3})(\text{m}^4 \text{s}^{-2})(\text{m}) = \text{J}$.

E3: Dimensionless line-energy kernel after length-scale normalization

$$\hat{H}_K = \sum_{i,j} \oint \oint \frac{d\hat{\mathbf{X}}_i \cdot d\hat{\mathbf{X}}_j}{\sqrt{\|\hat{\mathbf{X}}_i - \hat{\mathbf{X}}_j\|^2 + \hat{a}^2}} \quad (3)$$

Dimensional check: All hatted lengths are dimensionless, so the complete kernel is dimensionless.

E4: Self/mutual coherence matrix

$$C_{ij} = \frac{E_{ij}}{\sqrt{|E_{ii}E_{jj}|}} \quad (4)$$

Dimensional check: Energy divided by energy is dimensionless.

E5: Reduced conservative Lagrangian

$$L(q, \dot{q}) = \frac{1}{2} \dot{q}^T M \dot{q} - U(q) \quad (5)$$

Dimensional check: The blind reduced lane is nondimensionalized; after restoration both terms carry energy units.

E6: Reduced Hamiltonian from the Legendre transform

$$H(q, p) = \frac{1}{2} p^\top M^{-1} p + U(q), \quad p = M \dot{q} \quad (6)$$

Dimensional check: The blind reduced lane is dimensionless; after restoration H has energy units.

E7: Energy Hessian at a registered carrier geometry

$$K_H = \nabla_q^2 U(q) \big|_{q=0} \quad (7)$$

Dimensional check: Dimensionless in the normalized lane.

E8: First-order dynamical Jacobian

$$J = \begin{pmatrix} 0 & I \\ -M^{-1}K_H & -M^{-1}G \end{pmatrix} \quad (8)$$

Dimensional check: In normalized time J is dimensionless per unit normalized time; a chosen physical time scale restores s⁻¹.

E9: Spectral ratio with no protected target embedded

$$q_{21}(K) = \omega_2(K)/\omega_1(K) \quad (9)$$

Dimensional check: Ratio of equal-frequency units is dimensionless.

E10: SI mass prediction from the regularized Hamiltonian

$$M_K^{(\rho)} = H_K^{(\rho)} / c^2 \quad (10)$$

Dimensional check: J/(m² s⁻²)=kg.

E11: Three-dimensional kinetic-energy mass functional used as the physical reference object

$$M_K^{(3D)} = \frac{1}{2c^2} \int_{\mathbb{R}^3} \rho_K(\mathbf{x}) \|\mathbf{v}_K(\mathbf{x})\|^2 d^3x \quad (11)$$

Dimensional check: (kg m⁻³)(m² s⁻²)(m³)/(m² s⁻²)=kg.

E12: Core-tube quadrature kernel used as an independent reduced check on the line Hamiltonian

$$\hat{E}_{\text{tube}} = \frac{1}{2} \int_{\hat{\Omega}_{\text{tube}}} \|\hat{\mathbf{v}}(\hat{\mathbf{x}})\|^2 d^3\hat{x}, \quad E_{\text{tube}} = \frac{\rho \Gamma^2 r_c}{32\pi^2} \hat{E}_{\text{tube}} \quad (12)$$

Dimensional check: The hatted integral is dimensionless; rho Gamma² r_c has units J.

Registered code-to-mathematics steps

1. **S1** [G1; equations E3]: Verify E010 release metadata, locate qualification ledgers, reload original carriers, and recheck raw/geometry hashes when available.
2. **S2** [G2; equations E1, E2, E3]: Run analytic/synthetic controls, SI scale checks, orientation/rigid-motion/resampling controls.
3. **S3** [G3; equations E3]: Compute length, reach-normalized ropelength proxy, curvature, torsion, writhe/linking, Bishop holonomy, and convergence statistics.
4. **S4** [G4; equations E3]: Evaluate held-out independent carrier groups without pooling distinct carriers as numerical refinement levels.
5. **S5** [G5; equations E2, E3]: Compare C++/pybind11 FP64 regularized energy matrix with Python/NumPy FP64 reference.
6. **S6** [G6; equations E2]: Record framework SYCL/DD32 screening evidence separately from certification authority.
7. **S7** [G7; equations E3]: Quantify cross-independence-family replication and pseudo-replication exclusions.
8. **S8** [G8; equations E2, E4, E11, E12]: Compute self/mutual energy matrices, coherence spectra, and an independent Bishop-frame core-tube kinetic-energy quadrature for a bounded topology subset.
9. **S9** [G10; equations E5]: Evaluate reduced Euler-Lagrange residuals around source-native geometries even when earlier source/geometry gates fail, carrying caveats forward.
10. **S10** [G11; equations E5, E6]: Evaluate Legendre-transform closure and reduced Hamiltonian conservation.
11. **S11** [G12; equations E7]: Construct finite-difference Hessians under Bishop-frame perturbations and test symmetry/refinement.
12. **S12** [G13; equations E8, E9]: Construct Jacobians, extract spectra, and compare oscillatory frequencies with direct reduced integration.
13. **S13** [G14; equations E9]: Measure topology stability of spectral ratios without accessing the protected historical target.
14. **S14** [G15; equations E3]: Cross-validate simple geometry surrogates against the finite-core energy kernel with topology-grouped holdouts.
15. **S15** [G16; equations E3, E9]: Compare preregistered energy-only and geometry-plus-spectrum surrogate families under a complexity penalty.
16. **S16** [G17; equations E3, E10]: Freeze a complete target-free topology prediction table and anonymous composite predictions before reveal.
17. **S17** [G18; equations E3]: Evaluate protected topology holdout generalization internal to the geometry/energy task; no observed mass targets are used.
18. **S18** [G19; equations E2, E10]: Restore SI scales using canonical SST constants in two density lanes and write target-free absolute predictions.
19. **S19** [G20; equations E2, E7, E8]: Stress core regularization, discretization, carrier selection, and perturbation amplitude; quantify sensitivity.

20. **S20** [G21; equations E3, E9]: Jackknife across provider/lineage groups and report whether findings replicate in independent evidence families.
21. **S21** [G22; equations E3, E5, E6, E8, E10]: Assign a blind multi-level classification from all recorded evidence while preserving diagnostic results from failed upstream scientific gates.

7 Algorithm

Carriers are reloaded from source-native bytes, independently resampled, reach-normalized, and evaluated on preregistered resolution/core ladders. Geometry, self/mutual line energy, an independent core-tube kinetic-energy quadrature, coherence, reduced-action residuals, Hessians and Jacobians are computed without protected targets. Topology-grouped model competition predicts the energy kernel rather than observed masses.

8 Sources

The primary runtime source is E010 PKLSA v0.3.x. Carrier-level hash verification and independence labels are rechecked at runtime. Synthetic controls are implementation evidence only.

9 Gates and diagnostic continuation

Only G0 is a hard framework prerequisite. Every scientific gate depends directly on G0 rather than on the preceding scientific gate. Consequently a scientific FAIL does not prevent later computation. A separate soft-dependency ledger records interpretive caveats. This distinction is intentional: evidence is retained while unsupported promotion is prevented.

10 Numerics

BASIC uses a small selected set; FULL scans admitted topology classes through the registered crossing limit; CERTIFY increases carrier and resolution ladders. Regularized double line integrals are evaluated in FP64. A separate tube-support velocity quadrature probes the three-dimensional kinetic-energy route on a bounded subset. Hessians use centered finite differences under small Bishop-frame perturbations.

11 Backends

Python/NumPy FP64 is REFERENCE. C++/pybind11 FP64 is CERTIFICATION and must satisfy relative-L2 $\leq 10^{-10}$. SYCL FP32 and DD32/FP32x2 remain SCREENING_ONLY under framework v1.0.6 and cannot certify the scientific result.

12 Blindness

Protected observed targets, protected human identities and protected historical numerical comparison values reside only under `private/`. The blind run freezes a complete topology prediction table first. Reveal requires the v1.0.6 output-integrity and nonced-commitment checks.

13 Statistics

Primary diagnostics are within-topology coefficient of variation, between/within spread ratio, independent-group replication fraction, topology-grouped confirmation/holdout log-RMSE, carrier jackknife sensitivity, core-law sensitivity and target-free spectral-ratio stability.

14 Results

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	1339920fa01288a4d9d2d1e2bfb10fd9a9ca6d89ce473dc1d9fba7b88dedb2af
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	66c33f8b95eaa123899268a2f63943931be0a833acba3693c225a8cb06dedd04

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen protocol intact and is the runtime self-consistent?
G1	PASS	Are E010/PKLSA sources discoverable and are evaluated carriers individually proven?
G2	PASS	Do analytic controls, SI dimensional scales, rigid-motion/orientation/resampling controls pass?
G3	FAIL	Are geometry descriptors and the regularized energy kernel numerically converged and stable?
G4	UNRESOLVED	Do preregistered held-out independence/provenance groups reproduce the discovery-agnostic results?
G5	UNRESOLVED	Does strict native C++ FP64 reproduce Python/NumPy FP64 for the regularized energy kernel?
G6	PASS	If available, does framework GPU/DD32 screening evidence satisfy its frozen screening criteria?
G7	FAIL	Does the result replicate across independent PKLSA source/provider families without bias?
G8	PASS	For multicomponent carriers, does self+mutual energy reconstruct total energy and yield correct observables?
G10	PASS	Does the reduced candidate Lagrangian exhibit acceptably small Euler-Lagrange residuals?
G11	PASS	Does the Legendre transform close and is the reduced Hamiltonian conserved to the required precision?
G12	PASS	Is the finite-difference energy Hessian symmetric/converged, with gradient information consistent?
G13	PASS	Does the derived Jacobian spectrum agree with direct reduced integration for resolution and stability?
G14	UNRESOLVED	Are target-free spectral ratios reproducible within topology across independent carrier sets?
G15	PASS	Which preregistered geometry-only surrogate best predicts the finite-core energy kernel?
G16	FAIL	Does adding coherence/spectral information improve protected energy-kernel generalization?
G17	PASS	Has a complete target-free topology prediction table, including anonymous registered carriers?
G18	PASS	Do the selected blind model families generalize to a protected topology holdout not used in training?
G19	PASS	Can absolute SI predictions be generated from the frozen dimensionless kernel in both directions?
G20	PASS	Are primary conclusions robust to core regularization, discretization, carrier jackknife, and topology?
G21	FAIL	Do primary observables survive provider/lineage-group jackknife and independent evaluation?
G22	FAIL	What is the blind evidence classification when all diagnostic gates are considered with equal weight?
G9	DEFERRED	After a completed blind FULL/CERTIFY run, does commitment-verified reveal preserve the blind evidence?

Backend manifest

```
{
  "cpp_certification": {
    "authority": "CERTIFICATION",
    "gate_status": "UNRESOLVED"
  },
  "diagnostic_continuation": true,
  "gpu_screening": {
    "authority": "SCREENING_ONLY",
    "gate_status": "PASS"
  },
}
```

```

"python_reference": {
  "actual_backend": "python-numpy-fp64",
  "authority": "REFERENCE",
  "precision": "float64"
},
"schema": "SST-BACKEND-MANIFEST-2",
"scientific_gate_dependencies": "soft; see DEPENDENCY_CAVEATS.json"
}

```

Source manifest

```

{
  "manifest_sha256": "09a2f1c5e157041d3d89324ad0f801cfa5480222e0ae53f2be75d48208b0585c",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E010",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "E010_release_metadata",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E010_pklsa_p",
      "provenance_family": "E010_PKLSA_v0.3.1",
      "registry_hint": {
        "catalog_id": "E010",
        "hits": []
      },
    },
    {
      "role": "discovery_authority",
      "sha256": "23e4ec18be96fe4e8dde736a46bc9ccf898991d1858ddf3999770c34eed4f34a",
      "source_id": "S_E010_RELEASE",
      "version": "v0.3.1"
    },
    {
      "catalog_id": "E010",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "E010_release_metadata",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E010_pklsa_p",
      "provenance_family": "E010_PKLSA_v0.3.1",
      "registry_hint": {
        "catalog_id": "E010",
        "hits": []
      },
    },
    {
      "role": "topology_campaign_index",
      "sha256": "4ca56301102944c993ff90878ac468647e90c578bfc66e2f3d397238465da5c5",
      "source_id": "S_E010_INDEX",
      "version": "v0.3.1"
    },
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "E010_implementation",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\E_pipelines\\E010_pklsa_p

```

```

    "provenance_family": "E010_PKLSA_v0.3.1",
    "registry_hint": {
      "catalog_id": "E010",
      "hits": []
    },
    "role": "source_native_geometry_parser",
    "sha256": "597d5e3b06b0a91884edd2a7ffa99b2c7160940c68c9da39addb0d0c32796d26",
    "source_id": "S_E010_LOADER",
    "version": "v0.3.1"
  },
  {
    "catalog_id": "A057",
    "evidence_class": "simulation",
    "exists": true,
    "independence_family": "historical_model_definition",
    "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A057_Master",
    "provenance_family": "VAM_to_SST_translation",
    "registry_hint": {
      "catalog_id": "A057",
      "hits": []
    },
    "role": "historical_model_structure_only",
    "sha256": "1acdaf7517d10ccab86de1c12c78212011b6b89e9edaa445fdd4942ed93e31ac",
    "source_id": "S_HISTORICAL_MODEL",
    "version": "v0.1.0"
  }
],
"workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}

```

15 Interpretation

A later diagnostic may remain numerically informative even when an earlier scientific gate fails. Such a result is explicitly downgraded by `DEPENDENCY_CAVEATS.json`; for example a Jacobian around a nonstationary carrier is a local numerical diagnostic rather than a certified stability spectrum. A mass claim requires the separately registered blind prediction and post-reveal comparison.

16 Reproducibility

Run `run_00_install.cmd`, verify/freeze the protocol, then execute BASIC, FULL or CERTIFY. The canonical Workbench root defaults to `C:\workspace\projects\SST-Workbench` and may be overridden by `SST_WORKBENCH_ROOT`.

References

- [1] H. K. Moffatt, 1969, “The degree of knottedness of tangled vortex lines,” *Journal of Fluid Mechanics* 35, 117–129. DOI: 10.1017/S0022112069000991.
- [2] P. G. Saffman, 1992, *Vortex Dynamics*, Cambridge University Press. DOI: 10.1017/CBO9780511624063.
- [3] R. L. Bishop, 1975, “There is More than One Way to Frame a Curve,” *American Mathematical Monthly* 82, 246–251. DOI: 10.1080/00029890.1975.11993807.

- [4] C. C. Adams, M. Hildebrand, and J. Weeks, 1991, “Hyperbolic invariants of knots and links,” *Transactions of the American Mathematical Society* 326, 1–56. DOI: 10.1090/S0002-9947-1991-0994161-2.
- [5] H. von Helmholtz, 1858, “Über Integrale der hydrodynamischen Gleichungen, welche den Wirbelbewegungen entsprechen,” *Journal für die reine und angewandte Mathematik* 55, 25–55. Permalink: eudml.org/doc/147531.